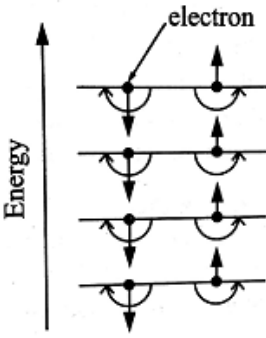
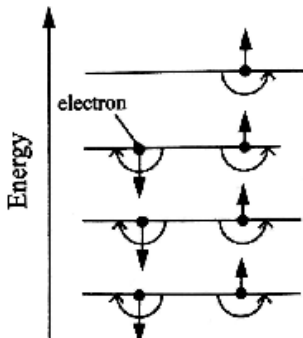
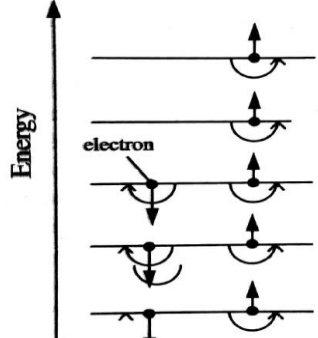
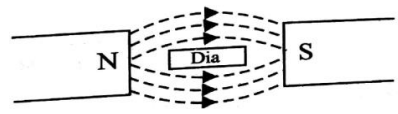
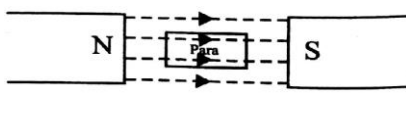
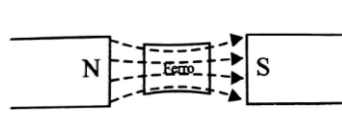


UNIT - III - MAGNETIC PROPERTIES OF MATERIALS

- Types of Magnetic materials
- Domain theory of Ferro magnetism
- Hysteresis
- Soft and Hard magnetic materials
- Ferrimagnetism or Ferrites
- Magnetic Storage Devices
- Magnetic Hard Disk Drive (HDD) with GMR sensor
- TMR sensor and Spintronics

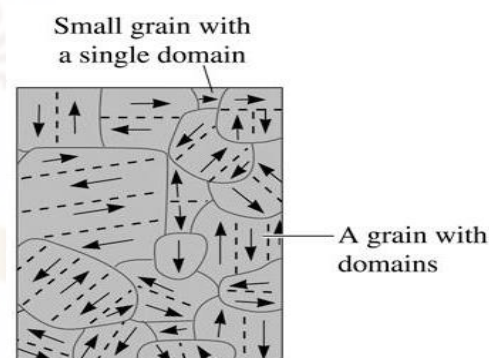
1. Types of Magnetic materials

S.No	Dia-magnetic material	Para-magnetic material	Ferro magnetic material
1	There are equal numbers of electron spins which are randomly oriented and hence the net magnetic moment is zero.	There are unequal number of electron spins and hence there exists a permanent magnetic moment.	There will be large number of unequal electron spins and hence there exists enormous amount of permanent magnetic moment
2			
3	When the external magnetic field is applied, the electrons will align perpendicular to the field direction.	When the external magnetic field is applied, the electrons will align parallel to the field direction.	When the external magnetic field is applied, the electrons which are already aligned parallel will reorient itself along the field direction.
4	When the material is placed in the magnetic field, the magnetic flux lines are repelled away from the material.	When the material is placed in the magnetic field, the magnetic flux lines pass through the material.	When the material is placed in the magnetic field, the magnetic flux lines are highly attracted towards the centre of the material.
5			

6	The susceptibility and intensity of magnetisation are negative.	The susceptibility and intensity of magnetisation are positive and small.	The susceptibility and intensity of magnetisation are positive and large.
7	The susceptibility is independent of temperature.	The susceptibility varies inversely with the absolute temperature.	The susceptibility depends upon the temperature.
8	Permeability is less than 1.	Permeability is greater than 1.	Permeability is very much greater than 1.
9	Weak magnets	Strong magnets	Very strong magnets
10	When the temperature is less than the critical temperature, the diamagnetism suddenly disappears and becomes a normal material.	When the temperature of the material is less than the curie temperature, para magnetic material is converted into diamagnetic material.	When the temperature of the material is greater than the Curie temperature, the ferromagnetic material is converted into paramagnetic material.
11	Examples: Gold, Antimony, Bismuth, Water, Silicon, Hydrogen, Germanium.	Examples: Platinum, Chromium, Aluminium, Copper sulphate, etc.,	Examples: Iron, Nickel, Cobalt, Steel, etc.

2. Domain Theory of Ferromagnetism

According to Weiss hypothesis, a single crystal of ferromagnetic material is divided into large number of well-defined small regions called domains. These domains have spontaneous magnetization due to the parallel alignment of spin magnetic moments in each domain. But the direction of spontaneous magnetization varies from domain to domain and are oriented in such a way that the net magnetization of the specimen is zero.

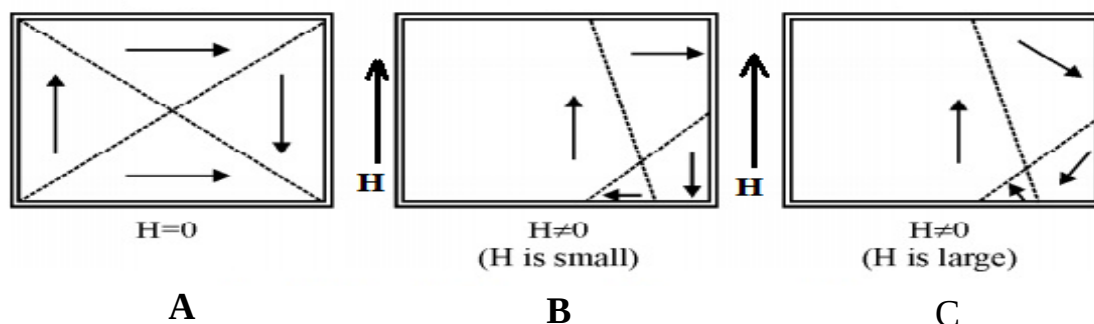


When the magnetic field is applied, the magnetization occurs in the specimen by two ways,

1. By the movement of domain walls
2. By the rotation of domain walls

By the movement of domain walls

The movement of domain walls takes place in weak magnetic fields. Due to this weak field applied to the specimen the magnetic moment increases and hence the boundary of domains are displaced, so that the volume of the domain changes as shown in the Figure B.



By the rotation of domain walls

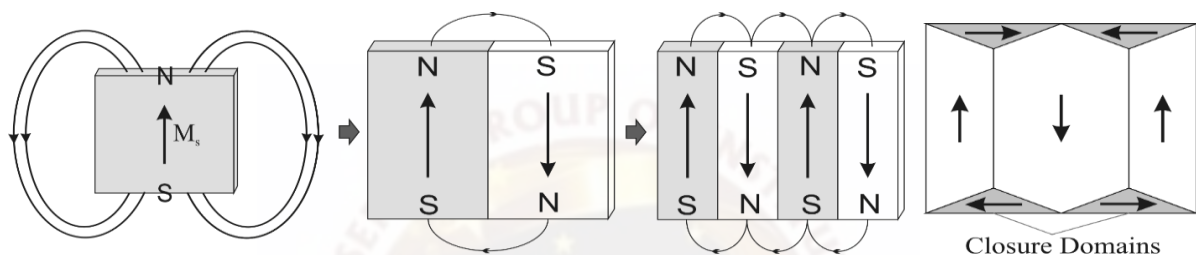
The rotation of domain walls takes place in strong magnetic fields. When the external field is high then the magnetization change by means of rotation of the direction of magnetization towards the direction of the applied field as shown in Figure C.

Types of Energy

The domain in ferromagnetic solid is understandable from the thermodynamical principle. (i.e.,) in equilibrium the total energy of the system is minimum. For this, first we consider the total energy of the domain structure and then how it is minimized. The total energy of the domain comprises the sum of following energies.

i) Exchange energy

The interaction energy which makes the adjacent dipoles to align themselves is known as exchange energy or magnetic field energy. In other words, it is the energy required in



assembling the atomic magnets into a single domain and this work done is stored as potential energy. The magnetic energy can be reduced by dividing the specimen into two domains as shown in above Figure.

The process of subdivision may be carried further, until the reduction of magnetic energy is less than the increase in energy to form another domain and its boundary. This boundary is called as domain wall or Bloch wall.

ii) Anisotropy energy

In ferromagnetic crystals there are two directions of magnetization.

a) Easy direction b) Hard direction

Weak field is applied in easy direction and strong field along hard direction. For ex: In iron [100] is easy direction, [110] medium direction and [111] is hard direction.

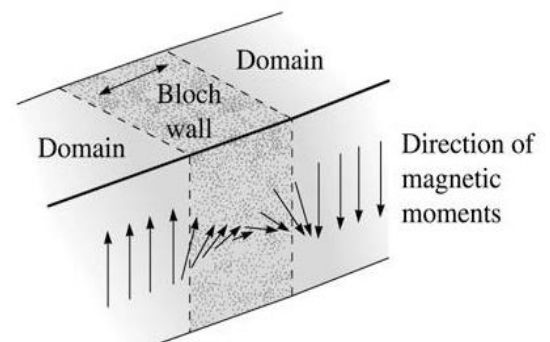
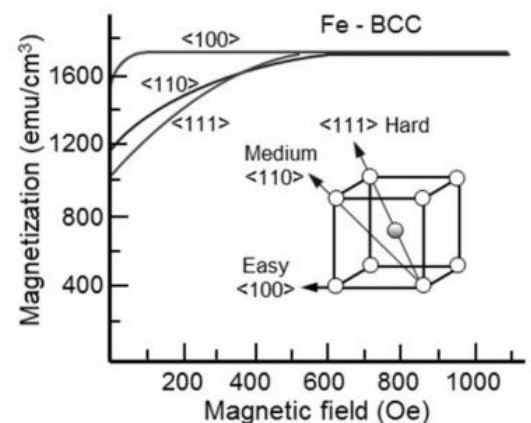
The excess of energy required to magnetise the specimen along hard direction over that required to magnetise the specimen along the easy direction is called crystalline anisotropy energy.

iii) Domain wall energy

Bloch wall is a transition layer which separates the adjacent domain magnetized in different directions.

Based on the spin alignments, two types of Bloch walls may arise namely i) Thick wall, ii) Thin wall.

When the spins at the boundary are misaligned and the spin changes gradually, then it leads to thick wall



(misalignment associated with exchange energy). In thin wall, the spins at the boundary changes abruptly, then the anisotropic energy becomes very less.

iv) Magnetostrictive energy

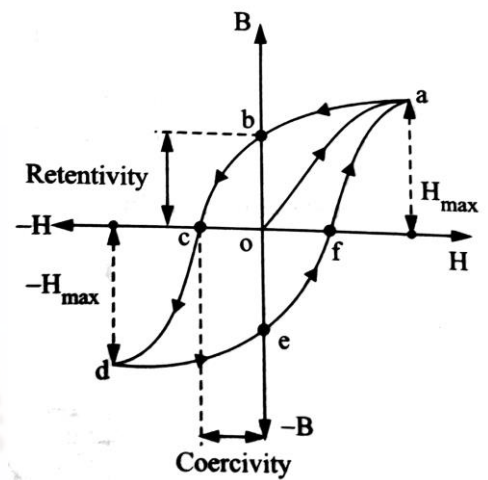
When a ferromagnetic material is suddenly magnetized it experiences a slight change in its length, which is due to the rearrangement of domains inside the material. This is called magnetostriction effect.

The magnetostrictive energy is the energy due the mechanical stresses generated by domain rotation.

Hysteresis

When ferromagnetic materials like Fe, Co, and Ni are made to undergo through a cycle of magnetization, the variation of magnetic induction (B) with respect to the magnetizing field (H) can be represented by a closed hysteresis loop or curve. i.e., it refers to the lagging of magnetic induction behind the magnetizing field.

If a magnetizing field H is applied to a ferromagnetic material and if H is increased to $H_{\max}$ the material acquires the magnetism. So the magnetic induction also increases, represented by 'oa' in the Fig. Now if the magnetic field is decreased from $H_{\max}$ to zero, the magnetic induction will not fall rapidly to zero, but falls to 'b' rather than zero. This shows that even when the applied field is zero (or) removed, the material still acquires some magnetic induction 'ob' which is so called residual magnetization (or) retentivity.



Now to remove this retentivity the magnetic field strength is reversed and increased to ' $-H_{\max}$ ' represented as 'oc' so called coercivity and hence we get the curve 'bcd'. Then the reverse field (-H) is reduced to zero and the curve 'de' is obtained, and by further increasing H to $H_{\max}$, the curve 'efa' is obtained.

Explanation of Hysteresis on the basis of domains

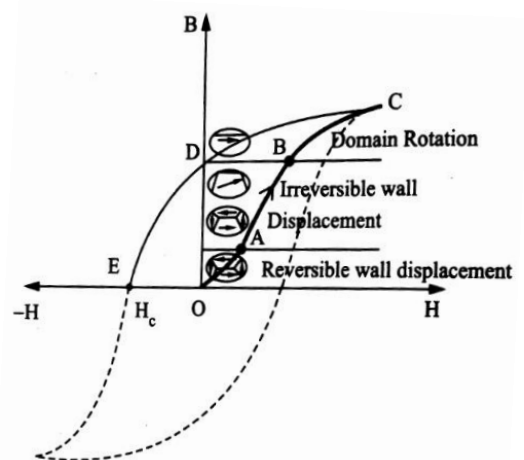
We know when the ferromagnetic material is subjected to external field, there is an increase in the value of the resultant magnetic moment due to two process,

- (i) The movement of domain walls
- (ii) Rotation of domain walls

When a small external field is applied, the domain walls are displaced slightly in the easy direction of magnetization. This gives rise to small magnetization corresponding to the initial portion of the hysteresis curve OA.

Now, if the applied field is removed, then the domains returns to its original state, and is known as reversible domains.

When the field is increased, large numbers of domains contribute to the magnetization and thus the magnetization increase rapidly with H.



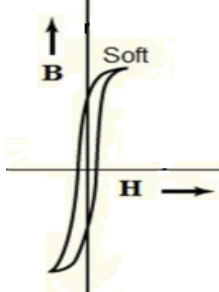
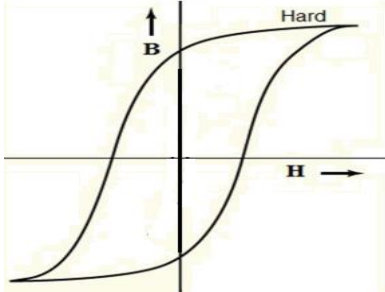
Now, when the field is removed, because of the displacement of domain wall to a very large distance, the domain boundaries do not come back to their original position. This process is indicated as AB in Fig. and these domains are called irreversible domains. At point 'B' all the domains have got magnetized along the easy direction.

Now, when the field is further increased, the domains start rotating along the filed direction and the anisotropic energy is stored in the hard direction, represented as BC.

Thus the specimen is said to attain the maximum magnetization. At this position, even after the removal of external field the material possess maximum magnetization, called residual magnetism represented by OD.

Actually after the removal of the external field, the specimen will try to attain the original configuration by the movement of Bloch wall. But this movement is stopped due to the presence of impurities, lattice imperfections etc. therefore to overcome this, a large amount of reverse magnetic field is applied to the specimen. The amount of energy spend to reduce the magnetization to zero is called as coercivity represented by OE.

3.Soft and hard magnetic materials

S.No	Soft	Hard
1.	They can be easily magnetized and demagnetized.	They cannot be easily magnetized and demagnetized.
2.	Have high susceptibility & permeability.	Low susceptibility & permeability.
3.	Magnetic energy stored is not high.	Magnetic energy stored is high.
4.	Low hysteresis loss, due to small hysteresis loop.	High hysteresis loss, due to large hysteresis loop.
5.	The eddy current loss is small due to its high resistivity.	The eddy current loss is high due to its low resistivity.
6.	Coercivity & Retentivity are small.	Coercivity & Retentivity are high.
7.	They are used to produce temporary magnets	They are used to produce permanent magnets
8.	Domain walls are easy to move.	The movement of domain wall difficult.
9.		
10.	Ex:- Ferrites, Garnets, Iron	Ex:- Al-Ni-Cu, Cu-Ni-Co, Carbon steel
11.	Used in switching devices, matrix storage computers	Microphones, Magnetic detector, Magnetic Separators.

4. Ferrimagnetism or Ferrites

Ferrimagnetic materials are also called as Ferrites. Ferrites are the modified structures of iron with no carbon and are composed of two or more sets of different transition metals. These materials have anti parallel magnetic moments of different magnitudes, giving rise to large magnetic moment in the presence of external magnetic field.

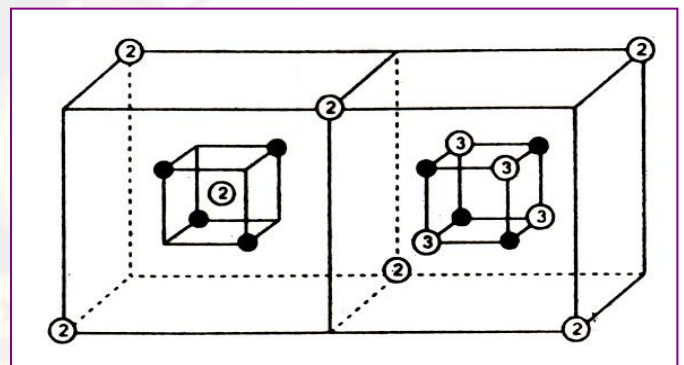
The general chemical formula of a ferrite molecule is $X^{2+}Fe_2^{3+}O_4^{2-}$, where X^{2+} represents a divalent metal ion such as Zn^{2+} , Fe^{2+} , Mg^{2+} , Mn^{2+} , Cd^{2+} etc., Ferrites crystallize in the form of a cubic structure. Each corner of a ferrite unit cell consists of a ferrite molecule.

Therefore, in a ferrite unit cell there are eight molecules. Therefore in a ferrite unit cell, there are eight divalent metal ions, 16 ferric ions and 32 oxygen ions. If only the oxygen ions in ferrite crystal are considered, it is found that they constitute a close packed face centered cubic structure.

In these arrangement it is found that for every four O^{2-} ions there are 2 octahedral sites (surrounded by 6 O^{2-} ions) and one tetrahedral site (surrounded by 4 O^{2-} ions). The metal ions are distributed over these tetrahedral sites (A sites) and octahedral sites (B sites). Thus in ferrites the number of octahedral sites is twice the number of tetrahedral sites. Normally there are two types of structures in ferrites.

i) Regular spinel structure

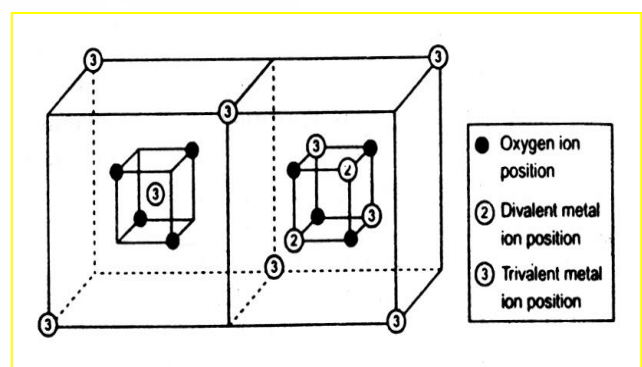
In this type, each divalent metal ion occupies 1 tetrahedral site and each trivalent metal ion occupies 1 octahedral site. Totally in an unit cell, there will be 8 tetrahedral (8 A) sites and 16 octahedral (16 B) sites. Hence, the sites A and B combined to form a regular spinel ferrite structures as shown in Fig.



The schematic representation of Zinc ferrite molecule as shown in Fig.

ii) Inverse spinel structure

In this type half of the B sites (8 sites) are occupied by divalent metal ions and the remaining half of the B sites (8 sites) and all the A sites are occupied by the trivalent metal ions, as shown in Fig.



Applications

- ❖ Ferrites are used in digital computers and data processing circuits. Ferrites are used to produce low frequency ultrasonic waves by magnetostriction principle.
- ❖ Ferrites are widely used in non-reciprocal microwave devices. Examples: Gyrator, Isolator and Circulator.
- ❖ Ferrites are also used in power limiting and Harmonic gyration devices.
- ❖ Ferrites can also be used in the design of ferromagnetic amplifiers of microwave signals.
- ❖ Ferrites are used as Audio and Video transformers.

- ❖ Ferrite core can be used as a bitable element.
- ❖ The rectangular shape ferrite cores can be used as a Magnetic shift register.
- ❖ Hard ferrites are used to make permanent magnets.
- ❖ The permanent magnets (hard ferrites) are used in instruments like Galvanometers, Ammeter, Voltmeter, Flex meters, Speedometers, Wattmeter, Compasses and Recorders.

Magnetic moment of a ferrite molecule (Ferrous ferrite)

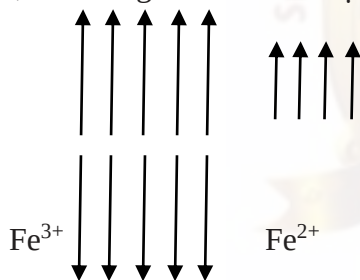
Saturation magnetization of a ferrite molecule can be calculated from the no. of unpaired spins of Fe^{2+} and Fe^{3+} .

Here we have two types of ions Fe^{2+} and Fe^{3+} .

- Fe^{2+} ions has six electrons in 3d shell out of 6 electrons two electrons are paired with each other and with 4 unpaired electrons. So, Fe^{2+} gives rise to 4 Bohr magneton
- Fe^{3+} has 5 electrons in 3d shell all are unpaired. We have two Fe^{3+} ions. Totally trivalent Fe^{3+} ions gives rise to 10 Bohr magnetons.

therefore, the total magnetization of Fe^{2+} and $\text{Fe}^{3+} = 14$ Bohr magneton is $14 \mu\text{B}$.

Theoretically we get $14 \mu\text{B}$ but experimentally the total magnetic moment got is only $4.08 \mu\text{B}$. The reason for this discrepancy is 'if all the spins are aligned parallel then we will get the total magnetization as $14 \mu\text{B}$. But in ferrites half of the magnetic spins of Fe^{3+} ions are parallel to one direction and the remaining half of Fe^{3+} ions are parallel in opposite direction and hence they cancel each other. So the net magnetic moment is only due to Fe^{2+} ions alone, i.e., total magnetization as $4 \mu\text{B}$.



Magnetic Storage devices

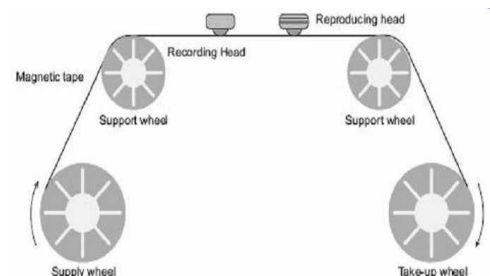
Magnetic Tape

Magnetic tape is one of the most popular storage medium for data. The tape is a plastic ribbon with metal oxide material coated on one side which can be magnetized, in this informations can be stored.

The electrical signal gives information about the stored data i.e., logic '1' or logic '0' states on the tape. In this way, the digital information stored is read out.

Advantages

- Magnetic tape is easy to handle and it is portable
- It is more compact and easier to handle
- Storing data on tapes is considerably cheaper
- Tapes have large storage capacities, ranging from a few hundred kilobytes to several gigabytes



Disadvantages

- It is a sequential access memory and hence access time is more
- Application
- They are generally used only for long term storage and backup.
- Tapes are also used for transporting large amount of data.

Magnetic Disks

Magnetic disks are the most popular direct access storage devices. These Magnetic disk memories provide large storage capabilities and moderate operating speeds.

Types of Magnetic disks

1. Hard disk
2. Floppy disk

Magnetic Hard Disc

The hard disk is used for storing a large amount of information. This Disk is available in different size such as 3.5 inch, 5.25 inch and 8 inch.

The hard disk is completely sealed and it is protected from the dust particles.

Construction

It is made up of hard aluminium platters. The platter surface is carefully machined until it is flat or plane. The platter surface is coated with magnetic material and is built into a box. It consists of a number of magnetic disks (2-5) or aluminium platters. All these platters are packed together and they are mounted on a common shaft.

This disk pack is placed in a drive mechanism hard disk drive. The drive mechanism drives the disk pack with the spindle. The data is written or read by R/W heads is the horizontal sensing arms by moving in and out between the platters with the precaution that the R/W head doesn't touches the surface instead it flyover the disk surface by a fraction of a milli meter.

The disk packs, read/write heads and the access mechanism are sealed in an air tight dust free container.

Advantages

1. It has very large storage capacity and more files can be permanently stored.
2. Very high speed in retrieving data.
3. The data can be accessed very large number of times without degradation.
4. This is prevented from dust particles, since they are sealed in the special chamber.

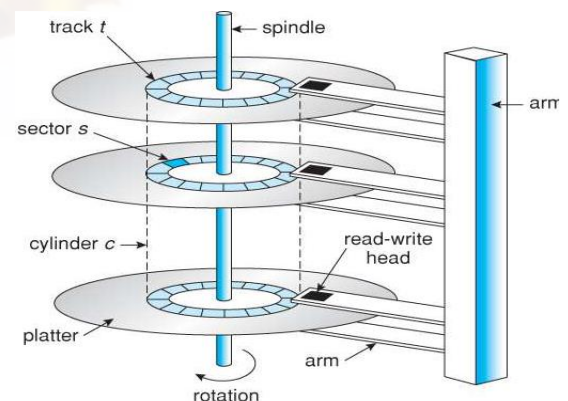
Disadvantages

1. Hard disks are not easily portable.
2. High cost and heavy loss for data.

Application

Hard disk is common secondary storage device for all types of computers.

The storage capacity of a hard disk drive may be ranking from 20 MB to 4 GB.



Magneto resistance

Some metallic materials show a large change in resistance on the application of magnetic field. This effect is called Magneto resistance (MR).

Magnetic Hard Disk Drive (HDD) with Giant Magneto Resistance (GMR) sensor

Principle

In hard disk drives, the binary data in terms of 0's and 1's are stored by inducing magnetic moment in a thin magnetic layer and GMR effect is used as the principle to read the data in HDD.

Zero represents missing transition and one represents transition in the medium.

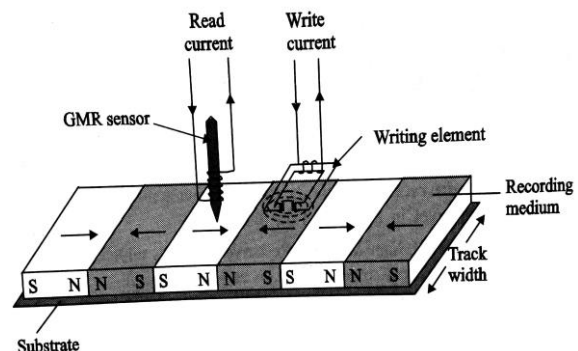
Construction

The HDD consists of recording medium made up of a thin layer of magnetic garnet grown over the substrate. The GMR sensor, which is made up of ferrites and antiferro magnetic materials is used as reading element. The writing element is made up of inductive magnetic transducer.

Working

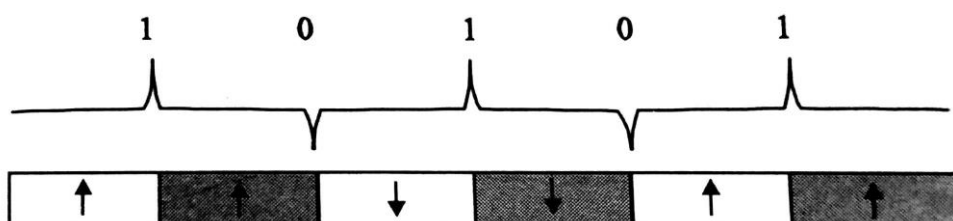
Writing/Storing

1. Initially the current is passed through the writing element and a magnetic field is induced in between the gap of the inductive magnetic transducer.
2. During writing the amplitude of current is kept constant, and the direction of the current is reverse.
3. So the magnetization orientation is reversed, from south to north.
4. When the induced magnetic field is greater than the coercivity of the recording media, the data is recorded in the form of 1.
5. 1 is stored in the recording medium.



Reading/Retrieving

6. When the GMR sensor is made to near the recorded medium, the resistance varies, with respect to the orientation of the magnetic moments.
7. When the layers are magnetized in parallel manner, the resistance is minimum, so maximum current flows through the sensor, which represents the data as one.
8. When the layers are magnetized in antiparallel manner, the resistance is maximum and so minimum current flows through the sensor, which represents the data as zero.
9. Therefore with help of the reading current, zeros and ones can be retrieved from the HDD.



Advantages

1. It has very large storage capacity.
2. Data can be stored in terms of Terabytes.
3. Non diffusive and very sensitive in reading.

Disadvantages

1. HDD has bulkier form factor.
2. Consume more power and will damage.
3. HDD may be corrupted sometime due to thermal radiation.

Applications

1. Used in coding and signal processing units.
2. Many engineering fields such as control systems, nano electronics etc.

SPINTRONICS

Spintronics is the study of the electron spin which transmits information. It combines magnetism and electronics also it plays vital role in solid state devices.

Principle

Spintronics works on the principle of the spin degree of freedom of electrons and holes which interact with their orbital moments.

Types of Spintronics

Metal-based spintronics: Using Fe, Co, Ni (Ferromagnetic devices)

Semiconductor based spintronics: Using Dilute Magnetic Semiconductor (DMS)

Working

Every electron has two stages, one spin up and spin down. In other words, electrons rotate either clockwise or anti-clockwise. However, in the case of ordinary materials, these electrons spin up and down and cancel themselves out. That is why ferromagnetic metal is the best one because it has the same aligned spin up in the electrons that are called domains.

These majority-up and majority-down domains are randomly scattered and with an externally applied magnetic field that will line up the domains in the direction of the electric field.

Advantages

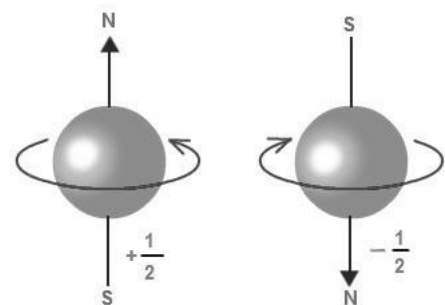
- ❖ Large storage capacity
- ❖ Faster devices
- ❖ Low power consumption
- ❖ No electric current required

Disadvantages

- ❖ Controlling the spin for distance is hard
- ❖ Difficult to inject and measure spin in silicon

Applications

- It is used to compress massive amounts of data into a small area, ie, 1 TB data can be stored on a single-sided 3.5" diameter disc.
- Used in magnetic random access memory (MRAM).
- In automotive industry due to its extremely small size, high sensitivity and low cost magnetic sensors.
- Missile guidance.
- Position and motion sensing in gaming.

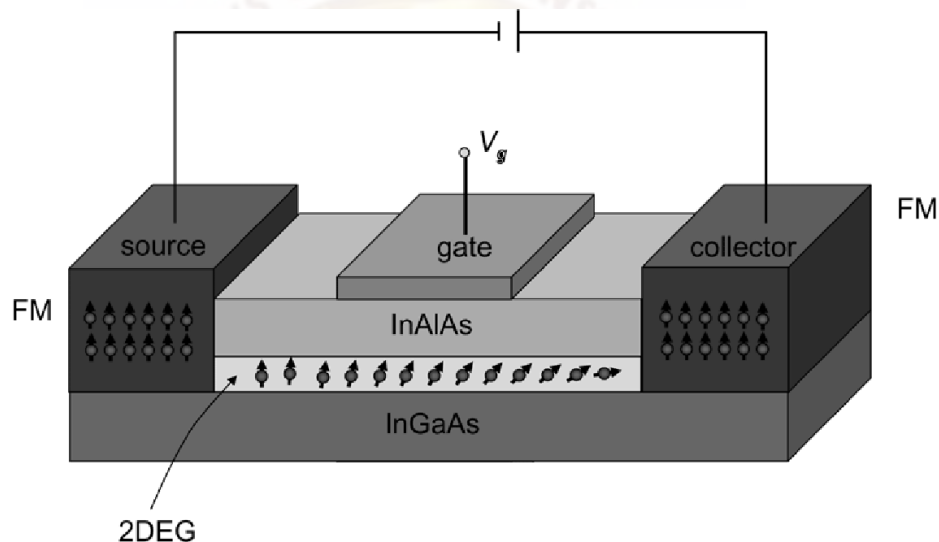


- To detect Cancer.
- Key hole surgery and post operative care.
- Used in the semiconductor industry to manufacture different types of transistors, lasers and integrated magnetic sensors.

SPINTRONICS DEVICES

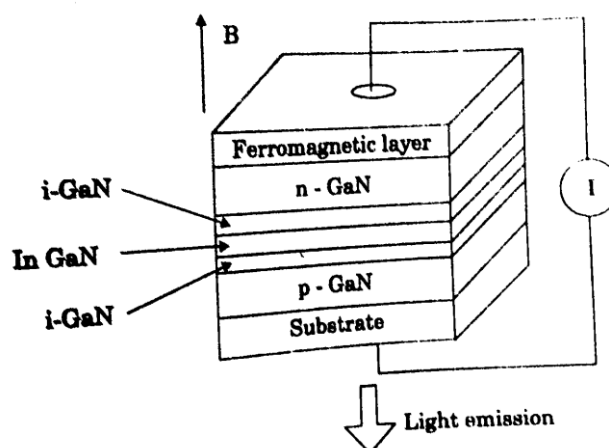
Spin transistor:

The Spin Field Effect Transistor consists of a source and a drain that are made of ferromagnetic materials. The basic idea of a spin transistor is to control the spin orientation by applying a gate voltage. A spin field-effect transistor (FETs) consists of ferromagnetic source and drain electrodes, a semiconductor channel that contains a layer of electrons, and a gate electrode attached to the semiconductor. The spin-polarised electrons are injected from the source electrode. The spin precession angle controls the flow of current. The gate electrode controls the rotation of the electron spin after entering the semiconductor channel. The success of these transistors depends on efficient injection of spin currents from a ferromagnetic metal to a semiconductor.



Spin LED

In the spin LEDs the emitted light possess a polarization dependent on the spin polarization of the charge carriers involved in the recombination. Due to this spin-dependent polarization, a device such as a spin LED can be used to produce light of a specific polarization. Additionally, spin LEDs can be used in the study and development of other spintronics devices.



TUNNEL MAGNETORESISTANCE(TMR) SENSOR

TMR sensor element is a new magnetic sensor technology that is less susceptible to temperature changes, has extremely high magnetic sensitivity and high resistance.

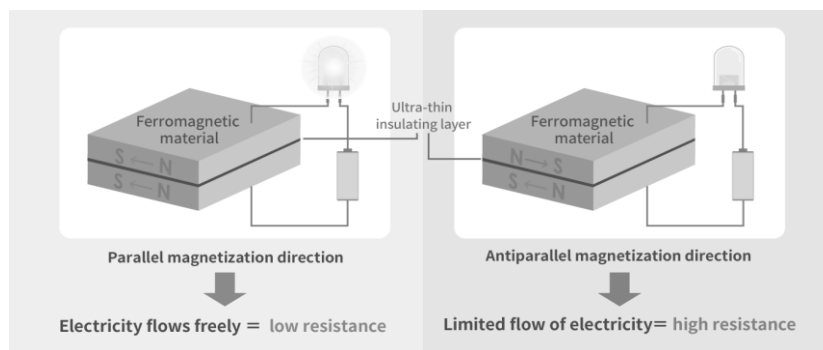
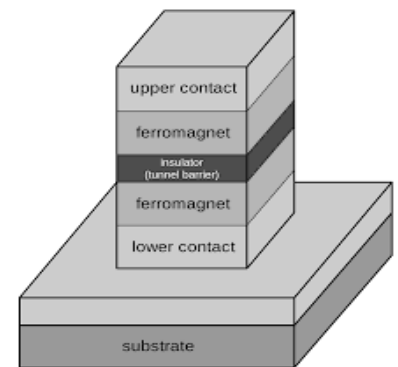
Tunnel magnetoresistance (TMR) is a magneto resistive effect that occurs in a magnetic tunnel junction (MTJ), which is a component consisting of two ferromagnets separated by a thin insulator. If the insulating layer is thin enough (less than 2 nanometres), allows a tunnel current to flow from one ferromagnet into the other.

GMR and TMR devices have a basic common structure, namely, two ferromagnetic metal films separated magnetically by a nonmagnetic film. The difference between the structures of these devices is in the nonmagnetic spacer film which consists of a metal film (GMR) or an insulator film (TMR).

Principle

The basic principle of TMR is **spin-dependent tunneling** i.e., up-spin electrons can only tunnel into up-spin states, and down-spin electrons can tunnel to down-spin states alone.

A TMR sensor element is configured from an extremely thin nanometer level nonmagnetic insulation layer sandwiched between two ferromagnetic layers. Electrons tunnel from one ferromagnetic layer into the other via insulation layer. This is a quantum mechanical phenomenon. Resistance decreases when the magnetization direction of the two ferromagnetic materials is parallel and increases when it is antiparallel.



Advantages

- ❖ High accuracy, High stability, Less aging deterioration

Applications

- ❖ Acceleration pedal sensors.
- ❖ Steering angle sensors.
- ❖ Used in cars' steering columns to improve the accuracy of automated parking capabilities.
- ❖ Current sensing, magnetometry or position sensing.